Descarga de datos y análisis de calidad

BBDD

Ensembl is a genome browser for vertebrate genomes that supports research in comparative genomics, evolution, sequence variation and transcriptional regulation. Ensembl annotate genes, computes multiple alignments, predicts regulatory function and collects disease data. Ensembl tools include BLAST, BLAT, BioMart and the Variant Effect Predictor (VEP) for all supported species.

http://www.ensembl.org/index.html
ftp://ftp.ensembl.org/pub/

BBDD

NCBI

National Center for Biotechnology Information

Welcome to NCBI

The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.

https://www.ncbi.nlm.nih.gov/
ftp://ftp.ncbi.nlm.nih.gov/

BBDD

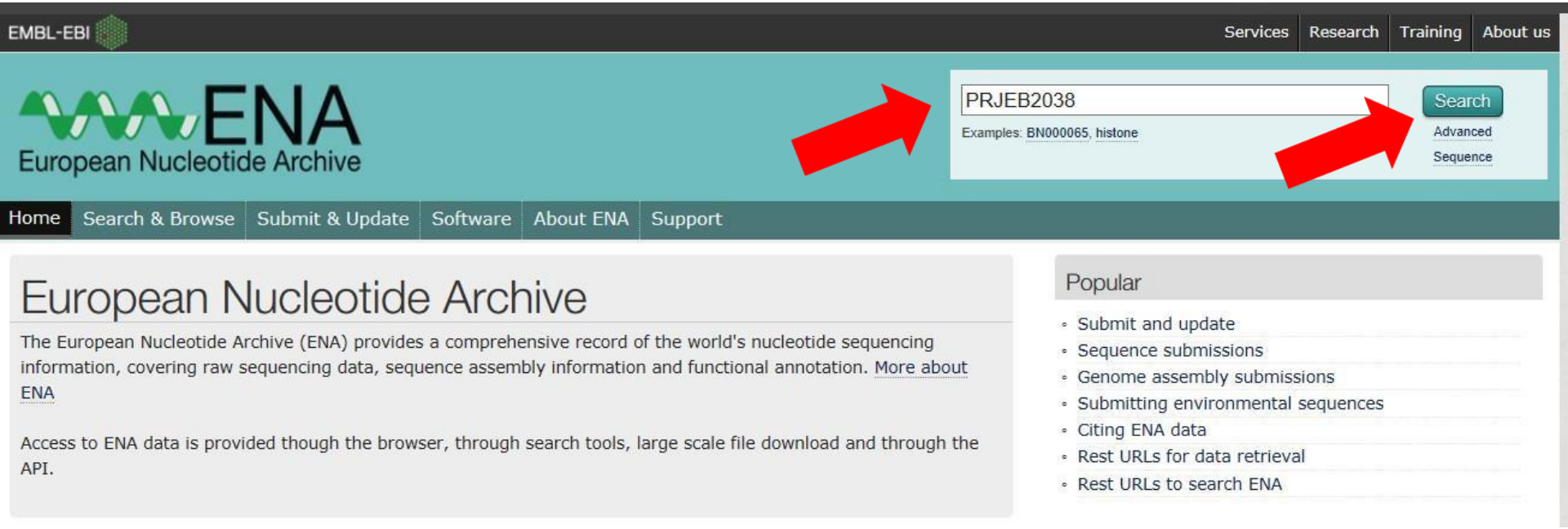
UNIVERSITY OF CALIFORNIA SANTA CRUZ UCSC Genome Browser

On June 22, 2000, UCSC and the other members of the International Human Genome Project consortium completed the first working draft of the human genome assembly, forever ensuring free public access to the genome and the information it contains. A few weeks later, on July 7, 2000, the newly assembled genome was released on the web at http://genome.ucsc.edu, along with the initial prototype of a graphical viewing tool, the UCSC Genome Browser. In the ensuing years, the website has grown to include a broad collection of vertebrate and model organism assemblies and annotations, along with a large suite of tools for viewing, analyzing and downloading data.

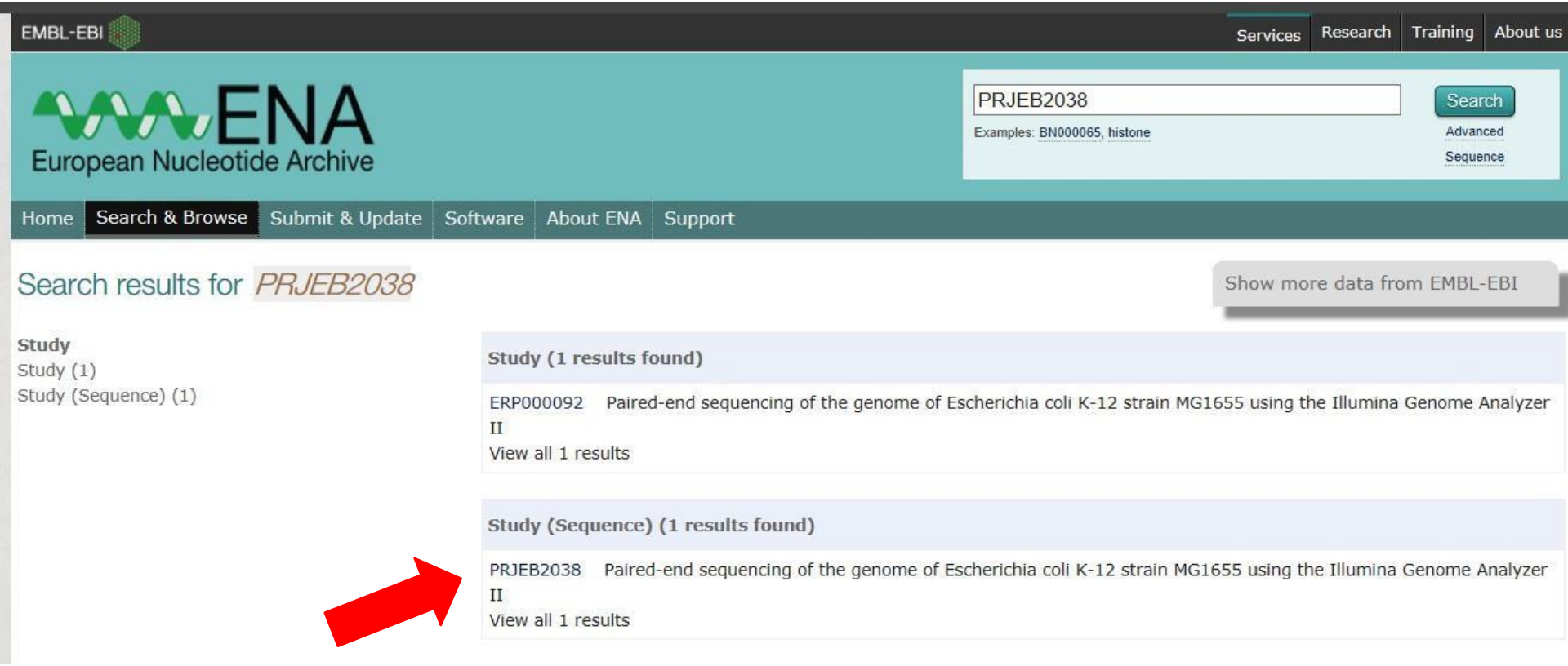
http://hgdownload.soe.ucsc.edu/downloads.html
ftp://hgdownload.cse.ucsc.edu/goldenPath/

Descarga de las lecturas usando la línea de comandos de Linux

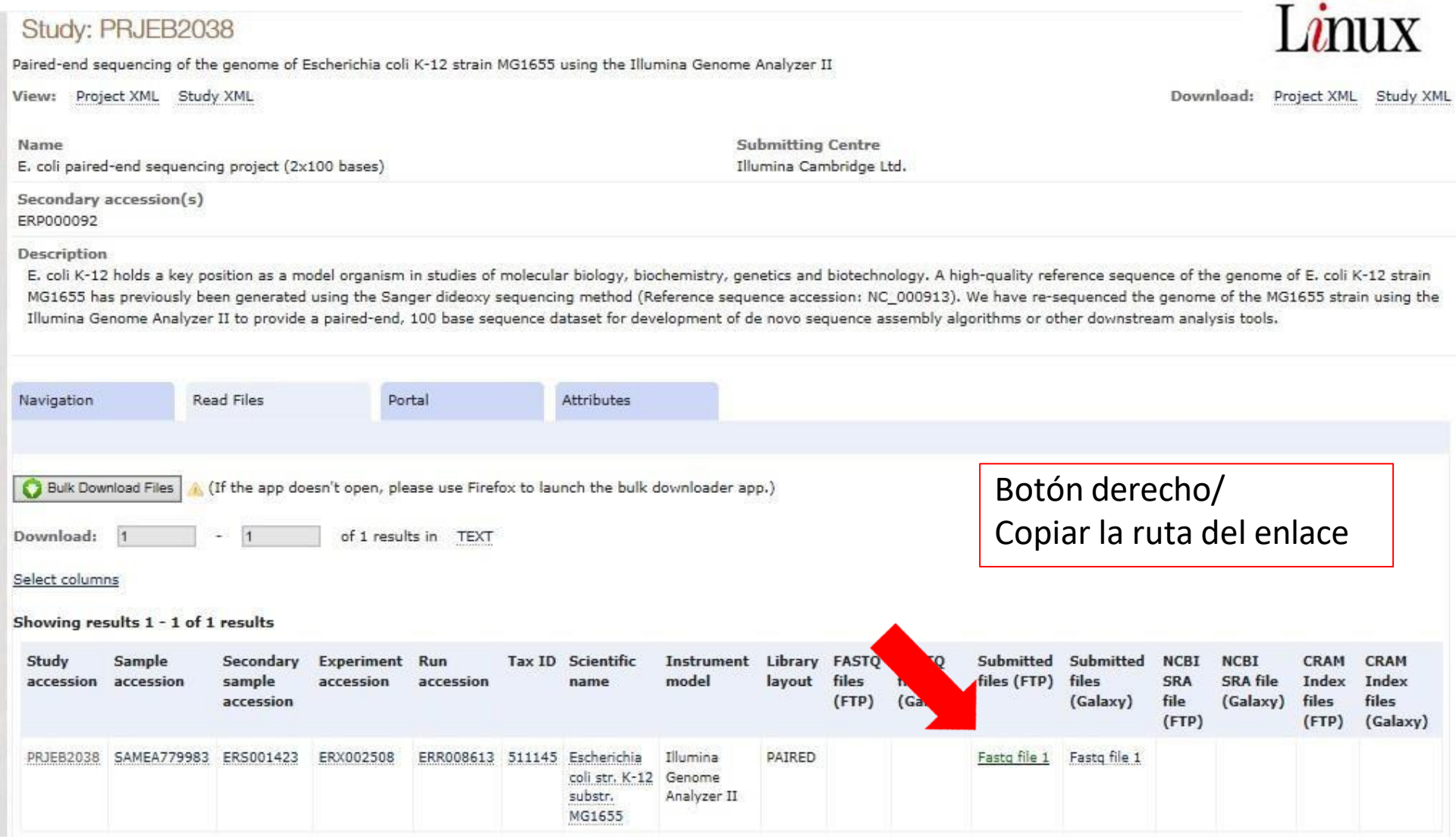
ENA Study PRJEB2038 → https://www.ebi.ac.uk/ena



Descarga de las lecturas usando la línea de comandos de Linux



Descarga de las lecturas usando la línea de comandos de Linux



Study: PRJEB2038

Paired-end sequencing of the genome of Escherichia coli K-12 strain MG1655 using the Illumina Genome Analyzer II

View: Project XML Study XML

Download: Project XML Study XML

Name
E. coli paired-end sequencing project (2x100 bases)

Submitting Centre
Illumina Cambridge Ltd.

Secondary accession(s)
ERP000092

Description
E. coli K-12 holds a key position as a model organism in studies of molecular biology, biochemistry, genetics and biotechnology. A high-quality reference sequence of the genome of E. coli K-12 strain MG1655 has previously been generated using the Sanger dideoxy sequencing method (Reference sequence accession: NC_000913). We have re-sequenced the genome of the MG1655 strain using the Illumina Genome Analyzer II to provide a paired-end, 100 base sequence dataset for development of de novo sequence assembly algorithms or other downstream analysis tools.

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Download: 1 - 1 of 1 results in TEXT

Select columns

Showing results 1 - 1 of 1 results

Study accession	Sample accession	Secondary sample accession	Experiment accession	Run accession	Tax ID	Scientific name	Instrument model	Library layout	FASTQ files (FTP)	FASTQ files (Galaxy)	Submitted files (FTP)	Submitted files (Galaxy)	NCBI SRA file (FTP)	NCBI SRA file (Galaxy)	CRAM Index files (FTP)	CRAM Index files (Galaxy)
PRJEB2038	SAMEA779983	ERS001423	ERX002508	ERR008613	511145	Escherichia coli str. K-12 substr. MG1655	Illumina Genome Analyzer II	PAIRED			Fastq file 1	Fastq file 1				

Botón derecho/
Copiar la ruta del enlace

Descarga de las lecturas usando la línea de comandos de Linux

- ✓ Abrir una terminal
- ✓ Crear un directorio para la práctica y cambiar a dicho directorio

```
$ mkdir ensamblaje
$ cd ensamblaje
```

- ✓ Descargar los archivos y descomprimir

```
$ wget
ftp://ftp.sra.ebi.ac.uk/vol1/ERA000/ERA000206/fastq
/200x100x100-081224_EAS20_0008_FC30TBBAAXX-6.tar.gz
$ tar -xzf  200x100x100-
081224_EAS20_0008_FC30TBBAAXX-6.tar.gz
```

- ✓ Mover los archivos a la carpeta actual y borrar la carpeta descomprimida

```
$ mv EAS20_8/* ./
$ rm -rf EAS20_8/
$ ls -al
```

Descarga de las lecturas usando la línea de comandos de Linux

El ensamblaje *de novo* es muy costoso computacionalmente, por lo que reduciremos el número de lecturas a 2 millones en cada archivo.

```
$ head -8000000 s_6_1.fastq > s_6_1_2M.fastq
$ head -8000000 s_6_2.fastq > s_6_2_2M.fastq
```

Ya tenemos las lecturas preparadas para poder ensamblarlas.

Descarga del genoma de referencia

http://www.ncbi.nlm.nih.gov/nuccore/49175990?report=fasta

NCBI Resources How To Sign in to NCBI

Nucleotide Nucleotide Search

Advanced Help

FASTA

Send to:

Escherichia coli str. K-12 substr. MG1655, complete genome

NCBI Reference Sequence: NC_000913.2

This sequence has been updated. See current version.

GenBank Graphics

```
>NC_000913.2 Escherichia coli str. K-12 substr. MG1655, complete genome
AGCTTTTCATTCTGACTGCAACGGGCAATATGTCTCTGTGTGGATTAAAAAAAGAGTGTCTGATAGCAGC
TTCTGAACTGGTTACCTGCCGTGAGTAAATTAAAATTTTATTGACTTAGGTCACTAAATACTTTAACCAA
TATAGGCATAGCGCACAGACAGATAAAAATTACAGAGTACACAACATCCATGAAACGCATTAGCACCACC
ATTACCACCACCATCACCATTACCACAGGTAACGGTGCGGGCTGACGCGTACAGGAAACACAGAAAAAAG
CCCGCACCTGACAGTGCGGGCTTTTTTTTTCGACCAAAGGTAACGAGGTAACAACCATGCGAGTGTTGAA
GTTCGGCGGTACATCAGTGGCAAATGCAGAACGTTTTCTGCGTGTTGCCGATATTCTGGAAAGCAATGCC
AGGCAGGGGCAGGTGGCCACCGTCCTCTCTGCCCCCGCCAAAATCACCAACCACCTGGTGGCGATGATTG
AAAAAACCATTAGCGGCCAGGATGCTTTACCCAATATCAGCGATGCCGAACGTATTTTTGCCGAACTTTT
GACGGGACTCGCCGCCGCCCAGCCGGGGTTCCCGCTGGCGCAATTGAAAACTTTCGTCGATCAGGAATTT
GCCCAAATAAAACATGTCCTGCATGGCATTAGTTTGTTGGGGCAGTGCCCGGATAGCATCAACGCTGCGC
TGATTTGCCGTGGCGAGAAAATGTCGATCGCCATTATGGCCGGCGTATTAGAAGCGCGCGGTCACAACGT
```

Choose Destination

- File
- Clipboard
- Collections
- Analysis Tool

Download 1 items.

Format FASTA

Show GI

Create File

enBank

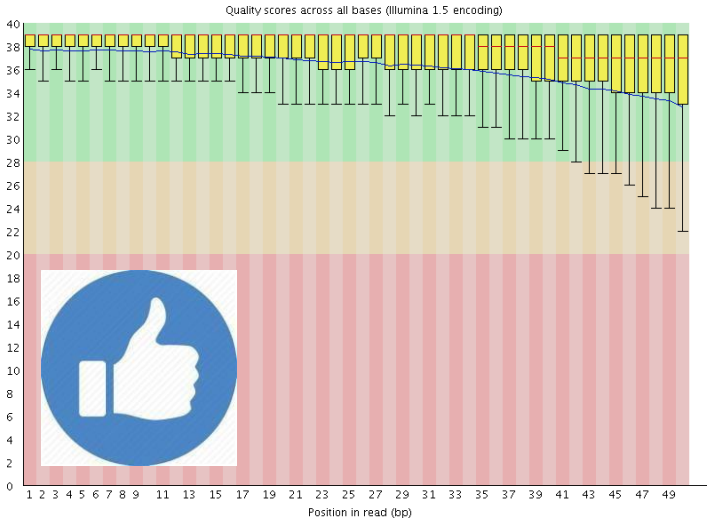
17 Service to America Award Finalist

Análisis de calidad: realizar siempre

Calidad de las lecturas: FastQC

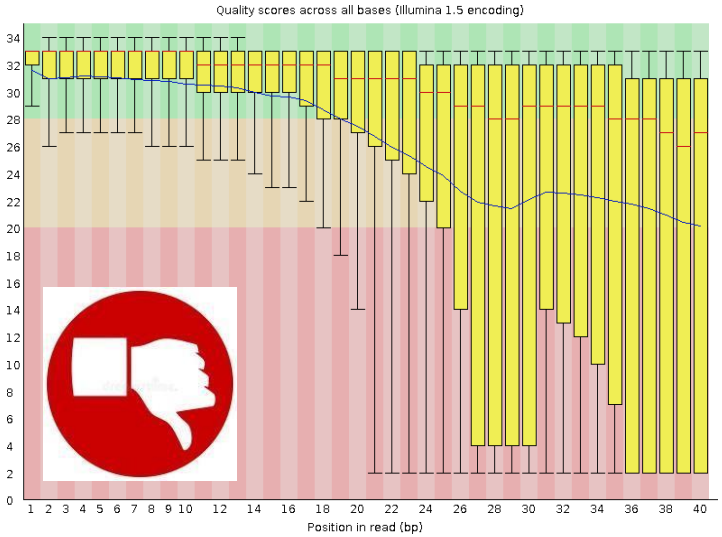
http://www.bioinformatics.babraham.ac.uk/projects/fastqc/

- ✓ Debemos ejecutar siempre tareas de control de la calidad para asegurarnos que los datos crudos tienen buena pinta y de que no hay problemas o sesgos en los datos que pueden afectar a su utilidad
- ✓ FastQC es una buena herramienta que permite ejecutar el análisis de calidad



¿Qué hago cuando FastQC considera que mis datos son de mala calidad?

- ✓ Mala calidad al final de la lectura -> recortar
- ✓ Fragmentos del adaptador en las lecturas -> recortar el adaptador



Calidad de las lecturas: FastQC

Calidad Phred

$$Q = -10\, log_{10}\, P$$

$$P = 10^{\frac{-Q}{10}}$$

$$Q = 10 \quad \Rightarrow \quad P = \frac{1}{10}$$

$$Q = 40 \quad \Rightarrow \quad P = \frac{1}{10000}$$

Calidad de las lecturas: FastQC

Código ASCII

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Calidad de las lecturas: FastQC

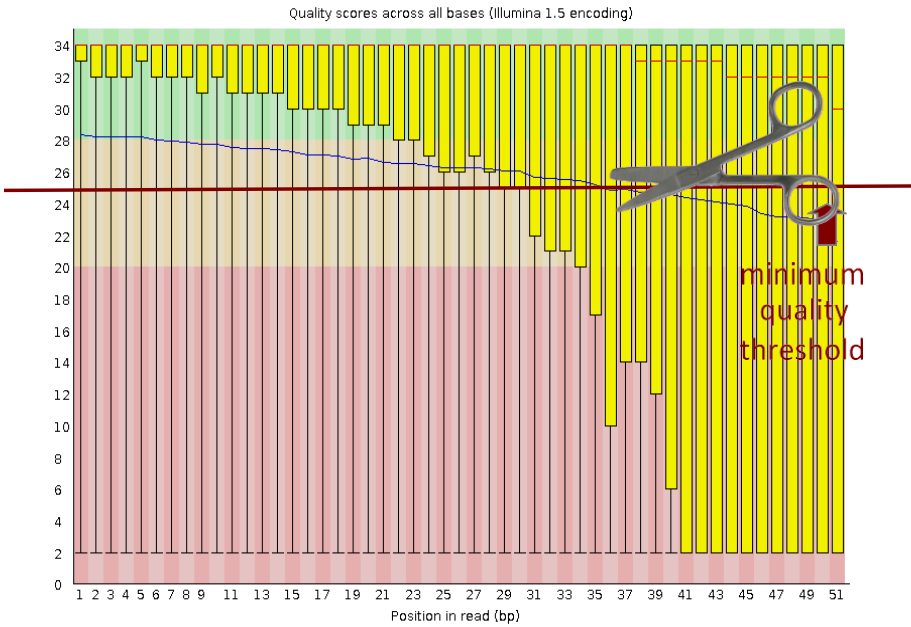
```
 SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....................................................
 ..........................XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX......................
 ...............................IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII......................
 ................................JJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ.....................
 LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL.....................................................
 !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
 |                         |    |        |                              |                     |
33                        59   64       73                            104                   126
 0........................26...31.......40
                          -5....0........9.............................40
                                0........9.............................40
                                   3.....9..............................41
 0.2......................26...31........41

S - Sanger        Phred+33,  raw reads typically (0, 40)
X - Solexa        Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64,  raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64,  raw reads typically (3, 41)
    with 0=unused, 1=unused, 2=Read Segment Quality Control Indicator (bold)
    (Note: See discussion above).
L - Illumina 1.8+ Phred+33,  raw reads typically (0, 41)
```

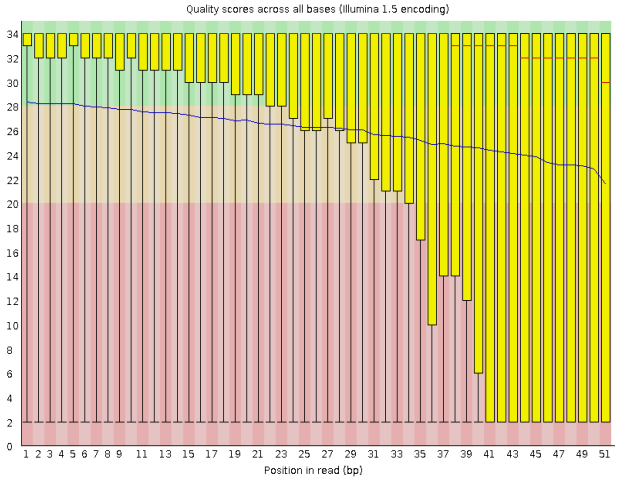
Filtrado y recorte

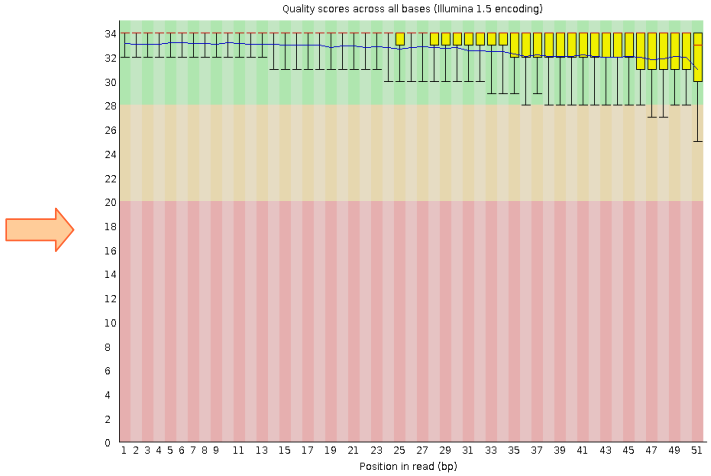
Filtrado de lecturas

- Calidad media, longitud
- Porcentaje de bases sobre una calidad predeterminada
- Recorte de adaptadores / zonas de mala calidad

Filtrar los datos de mala calidad puede mejorar nuestra confianza y el análisis posterior.







¡Esto es todo!

GRACIAS